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A systematic review of proactive approach for pedestrian safety at intersections

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ABSTRACT

Every human being is a pedestrian, and walking is the most traditional and often quite convenient mode of transportation. In recent years, walking has become exceptionally dangerous as it possesses risk of accidents, injuries and even fatalities. These accidents are not solely caused by vehicles but are largely due to pedestrian behaviour as well. This has warranted for a comprehensive study and understanding of pedestrian behaviours. The available state of the art pertaining to pedestrian safety approach can be broadly classified into reactive, proactive and perception. In this paper a systematic review pertaining to proactive approach is performed. The review procedure consists of selection criteria, article search scheme, article selection and data processing. Several factors associated with pedestrian personal attributes and site-specific parameters are found to have significant impact on pedestrian road-crossing decisions. Pedestrian behaviours are stochastic and have more freedom for their movement. Further, each intersection has unique characteristics and might require targeted engineering and behavioural interventions. Therefore, region and area-specific understanding is required, to suggest measures to improve pedestrian safety.

1. Introduction

The global population increase and economic prosperity have resulted in a rapid expansion of motor vehicle fleets and significant additions of road networks. This has put tremendous pressure on the safety aspects of road and traffic infrastructures. Road accidents are a major concern worldwide. There were an estimated 1.19 million road accident fatalities globally in 2021 (World Health Organization, 2023). The situation in developing and third-world countries is alarming. Data shows that 90 % of the traffic fatalities occurred in low and middle-income countries (World Health Organization, 2023). The share of fatalities in road accidents for various WHO regions are shown in Fig. 1.

Pedestrians and cyclists are considered the most vulnerable road users (VRUs). Most road accident victims are pedestrians, and their safety should be a top priority globally. Worldwide, pedestrian deaths account for 23 % of total fatalities. The African and the West Pacific Regions are the major contributors to pedestrian fatalities (World Health Organization, 2023). In India, pedestrians made

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up 19.5 % of the total fatalities in road accidents in 2022, an increase of 12.7 % compared to 2021 (Ministry of Road Transport and Highways (MoRTH). Govt. of India, 2023).

Intersections are critical locations where various traffic movements—such as turning, weaving, merging-diverging, and through traffic—interact, often leading to significant conflicts between vehicles and pedestrians (Haque et al., 2024). Owing to frequent and often unsafe pedestrian-vehicular interactions, pedestrian crashes are more likely to occur at intersections within shared spaces. A research report focusing on European countries reveal that intersections account for approximately 8 %–35 % of accidents, with a significant portion occurring at T and staggered intersections (European Commission, 2018). In Delhi about 31 % of accidents occurred at intersections resulting in 31.5 % of fatalities (Ministry of Road Transport and Highways (MoRTH). Govt. of India, 2023). Accidents involving pedestrians at intersections accounts for approximately 50 % and 60 % in China and Japan respectively (Marisamynathan & Vedagiri, 2018).

Over the past decades, extensive research has been conducted to understand the multifaceted nature of pedestrian safety. This body of work spans various disciplines, including behavioural science, traffic infrastructure design, traffic engineering and transport planning. Researchers have explored how pedestrians interact with their environment, the impact of urban design on safety, the effectiveness of traffic regulations, and the potential of technological innovations to prevent accidents. Despite these efforts, pedestrian safety remains a significant challenge, influenced by numerous factors such as demographic characteristics, environmental conditions and human behaviour.

Researchers have considered several factors related to pedestrians, drivers, road infrastructure, traffic conditions, accident data and built environment for assessment and improvement of pedestrian safety. Pedestrian demographics, socioeconomics, crossing behaviour, perception and road crossing decisions have been studied in past research. The keyword network map, presented in the studies related to the pedestrian safety (Fig. 2), highlights the most influencing factors considered in the past research. The colour is representing the number of occurrences of that factor in the previous studies.

Crash data and accident analysis are used extensively in developed countries for safety evaluation. Pedestrian facilities and road infrastructure are significant parameters in several pedestrian safety studies. Traffic related parameters such as speed and volume are also found to be crucial for pedestrian safety. Efforts and research by various stakeholders to understand pedestrian safety issues have started to yield positive results, as evidenced by a 5 % reduction in road accident fatalities from 2010 to 2021 (World Health Organization, 2023). Yet, this reduction is far below the target of reducing road accident fatalities by 50 % by 2030. The percent change in road accident fatalities in various WHO regions is shown in Fig. 3.

Pedestrian safety has emerged as a significant concern globally, with urbanization and increased vehicle ownership contributing to rising pedestrian accidents. Despite advances in vehicle safety technologies, pedestrian safety remains a pressing issue. The three primary approaches focussing on pedestrian safety are: a) Reactive Approach (RA); b) Proactive Approach (PA) and c) Perception Based Approach (PBA) (Angulo et al., 2023; Chandrappa et al., 2021; Mukherjee & Mitra, 2020c). The RA relies on historical crash data and accident records to identify high-risk locations and patterns in pedestrian safety issues. Interventions are developed based on past incidents, making this approach more suited to contexts where comprehensive accident data are readily available. RA is often effective in well-documented, data-rich environments but may have limitations in regions with underreported or inconsistent data. In contrast, the PA emphasizes identifying potential safety risks before accidents occur, using surrogate safety measures like pedestrian behaviour, compliance with signals, and near-miss observations. Proactive observation involves systematically monitoring pedestrian behaviour to identify risky crossing patterns before accidents occur. Methods such as video surveillance, direct observation, simulations and wearable sensors are widely used for proactive observations. This approach is especially valuable in areas where reliable crash data are scarce or incomplete, as it provides an alternative to data-intensive reactive measures. By focusing on early risk indicators, the PA aims to prevent accidents through pre-emptive interventions. PBA evaluates safety from the perspective of pedestrians, assessing their subjective feelings of risk, comfort, and difficulty in crossing intersections. The perception approach focuses on understanding and improving how pedestrians perceive their safety in various environments, which includes factors such as environmental design, traffic conditions, and social influences. This approach contributes to safer environments by addressing psychological and social aspects of pedestrian safety. It provides insight into how pedestrians experience intersection safety and complements both the reactive and proactive approaches by addressing factors that may not be apparent in observed behaviour or accident data. In this review proactive

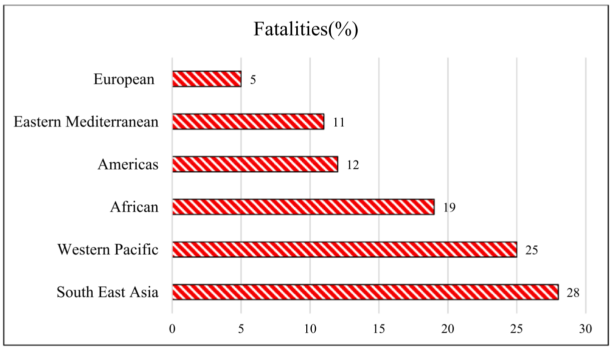

Fig. 1. Road accident fatalities for various WHO regions (World Health Organization, 2023).

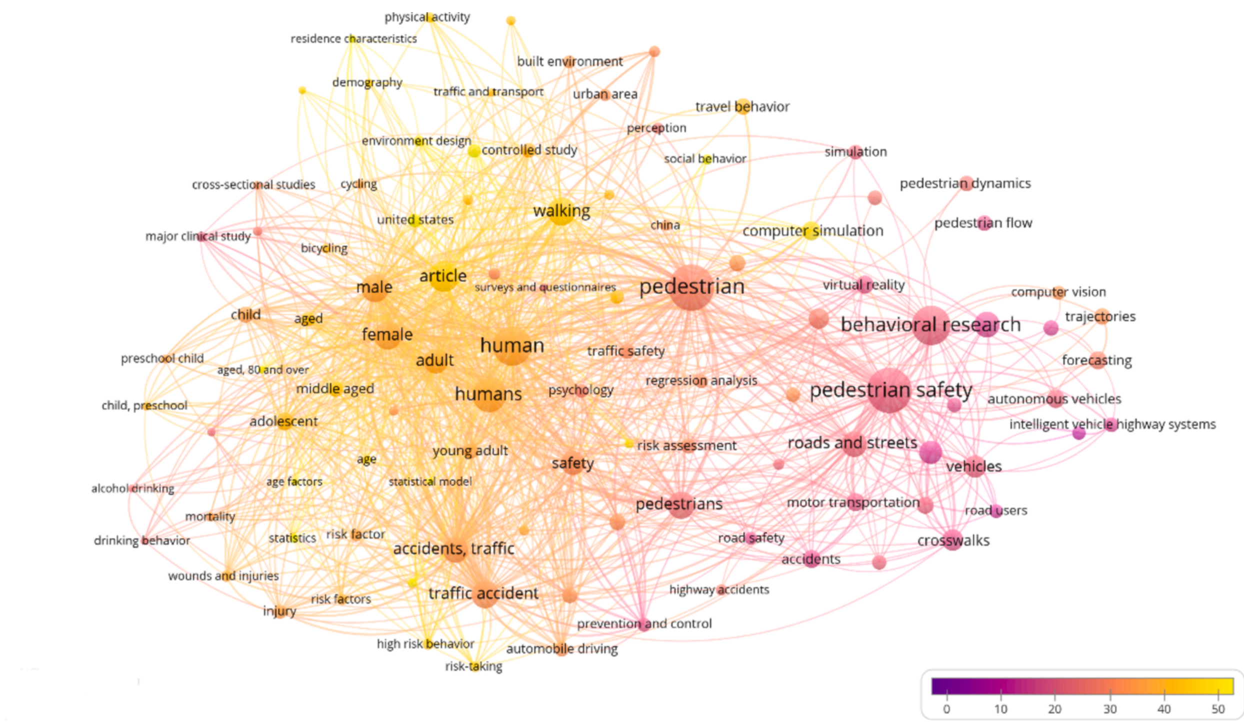

Fig. 2. Authors' keyword network map.

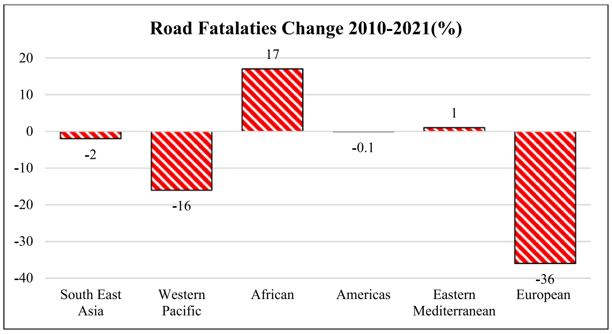

Fig. 3. Road accident fatalities change for various WHO regions (World Health Organization, 2023).

approach is explored in detail to draw insights and propose recommendations for future research and practical applications.

This review aims to provide a comprehensive exposition of the research conducted over the past two decades, focusing primarily on the proactive approach to enhancing pedestrian safety. By examining previous studies, this paper aims to identify various factors affecting pedestrian safety at intersections. Thus, the objectives of the current study are to:

i) Identify key behavioural factors influencing pedestrian safety: This objective focuses examining pedestrian demographic characteristics and behaviour patterns. **ii) Critical review of key intersection factors affecting pedestrian behaviour:** The study aims to critically review the effect of various intersection, traffic and built environment factors on pedestrian behaviour.

These objectives highlight a proactive approach to pedestrian safety, emphasizing risk anticipation and targeted preventive measures to make intersections safer and more pedestrian-friendly. In reviewing the literature, this paper addresses the following questions: i) How suitable and applicable are different pedestrian safety approaches across various economies? ii) What are the key factors influencing pedestrian behaviour and their impact on safety? and iii) How do factors related to traffic, road infrastructure, intersection and built environment contribute to enhance pedestrian safety. By addressing these questions, this review seeks to contribute to the ongoing discourse on pedestrian safety and provide a foundation for more effective interventions. Ultimately, the goal is to reduce pedestrian injuries and fatalities, making urban environments safer and more liveable for all road users.

2. Methodology

This study adopts a systematic approach to identify and analyse relevant literature as it has numerous advantages over conventional methods. The current work focuses on exploring the proactive approach for studying and assessing the pedestrian safety. Articles that have focused and covered the use of proactive approach for identification of various factors related to pedestrian safety from the year 2000 to 2023 are considered. To ensure a comprehensive understanding and avoid review bias, this study formulated a review procedure that includes selection criteria, article search scheme, article selection and data processing. The documents were processed following the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) extension for scoping reviews (Page et al., 2021).

2.1. Selection criteria

The selection criterion were established before choosing articles for the review of pedestrian safety studies. Peer-reviewed journal articles published between 2000 and 2023 that discuss pedestrian safety issues, particularly focusing on proactive approach were included. It is observed that there is little research from developing and third world economies, focussing on pedestrian safety. Therefore, articles from these developing countries were actively sought to fill this research gap. Articles published in languages other than English, as well as, reports, books, editorial documents and grey literature, are excluded from the review.

2.2. Article search scheme

A wide range of keywords focusing on the main theme of this review was used to explore the relevant literature. Keywords such as "pedestrian safety", "pedestrian behaviour", "intersection safety", "pedestrian accidents"," pedestrian violations" "pedestrian safety and risk perceptions", "proactive pedestrian safety" and "proactive measures in pedestrian safety" are included. Two databases – Scopus (managed by Elsevier) and Web of Science – WoS (managed by Clarivate) and search engine – Google Scholar (managed by Google) are used to retrieve the articles as per the search keywords. Initially Scopus and Web of Science databases are searched using specific search keywords. Subsequently, the screening of the reference lists of selected articles (both backward and forward screening) to identify any additional relevant work are carried out. Finally, to avoid the omission of any relevant literature in the initial search, an advanced search in Google Scholar was performed.

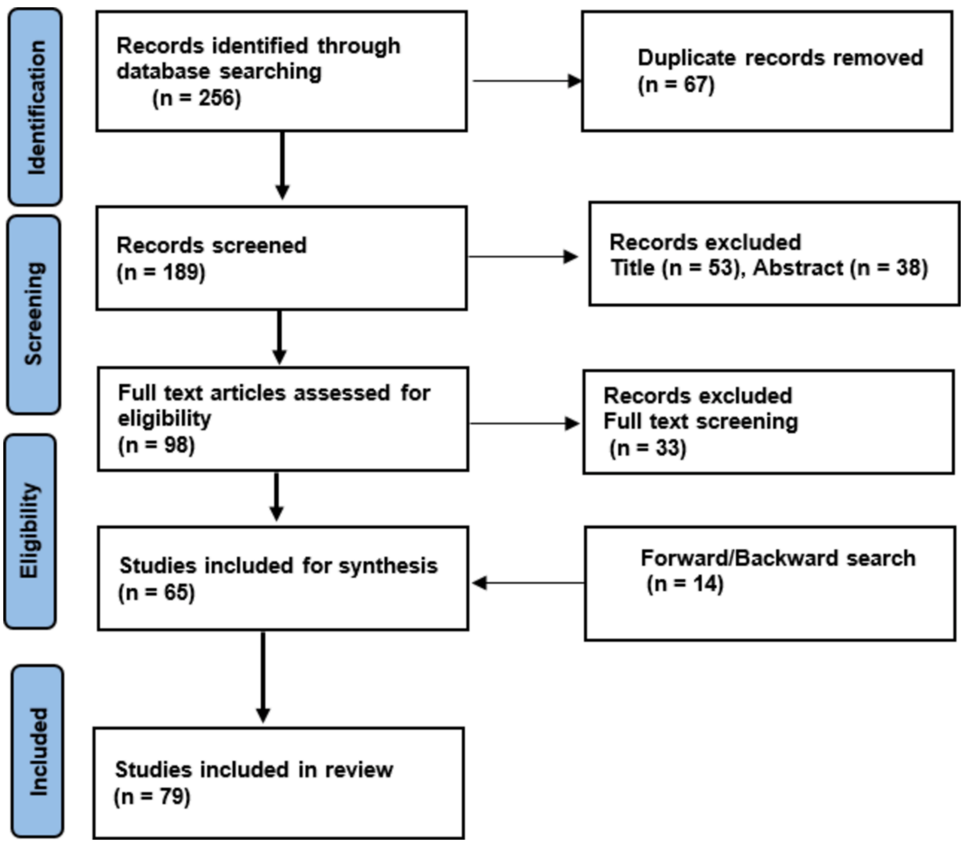

Fig. 4. Article screening process using PRISMA method.

2.3. Article selection

All the articles were collected and imported into Mendeley by utilizing the systematic search criteria and repetitions, if any are removed. The acquired documents are further processed through three stages of screening by using PRISMA method as shown in Fig. 4. In first stage, the titles of the articles were screened and those that did not meet the primary objective of this study were omitted. In second stage, the abstracts of the articles obtained after first stage were analysed to exclude those not relevant to the scope of this study. Finally, in third stage, full-text screening of articles was performed based on the defined selection criteria. Based on the above selection criteria, 79 articles were ultimately considered for this review.

2.4. Data processing

A spreadsheet was specifically designed to extract the required data from the selected manuscripts. Manuscript publication information such as journal name, author details, publication year, title and keywords was initially extracted. Subsequently, the research theme, and methodology pertaining to data collection, extraction, analysis and modelling were obtained. Information related to the size of the data and study location were also obtained. Finally, a detailed full-text review was conducted to get the information such as the significant factors considered for the study, research outcomes, critical discussions, recommendations, suggested interventions, research gap and scope for future works. The distribution of the studies by publication year and geographical area is shown in Fig. 5 and Fig. 6, respectively. The summary of the selected literature related to the proactive approach is given in Table 1.

3. Results and discussion

In this review the proactive approach for pedestrian safety is considered. So, the entire discussion in this section and sub-sections is primarily focused on the Proactive Approach (PA). Proactive approach for the assessment of pedestrian safety is also considered as the Surrogate Safety Measure (SSM). It is termed as proactive approach because it relies on anticipated potential risks rather on past events, such as accidents (Fitzpatrick et al., 2018). This approach primarily uses pedestrian behaviour and perception, field expert opinions, feedback and road safety audits (RSA) to identify and assess potential hazards and risky behaviours. The use of proactive approach has been suggested by researchers because of its cost-effectiveness (Kadali & Vedagiri, 2016) and added advantage that it does not wait for an accident to occur rather it tries to identify and neutralize potential risks and thus preventing an accident from happening. Reactive approach is extensively used in developed countries, but it has its own limitations with respect to crash data and application challenges, particularly in developing countries (Dandona et al., 2011). The various limitations pertaining to crash data as highlighted by various researchers with respect to developing countries are listed below:

i. **Underreporting:** Accidents are often underreported combined with deficient and incoherent records. Mostly fatal and major injuries are reported whereas minor injuries often go unreported (Dandona et al., 2008; Mohan et al., 2009; Singh et al., 2018).
ii. **Technical and Engineering Details:** Many times, police personnel are not trained enough to capture various crucial information such as road geometry, built environment (BE), nature and type of collision, traffic operation and control features and weather conditions. Sometimes socio-demographics of victims are also missing (Mukherjee & Mitra, 2020b).
iii. **Accuracy & Reliability:** Due to the lack of proper training the accuracy of police records for accidents is not much reliable in countries such as India (Bandyopadhyaya, 2015).
iv. **Hardship & Time Considerations:** Statistical analysis and modelling requires data for a larger period of time (3–5 Years). Obtaining accurate and reliable accidents data from police records is very tedious and time consuming. Further it is often not possible to collect sufficient data to suggest any counter measures (Kadali & Vedagiri, 2016; Mukherjee & Mitra, 2020c).

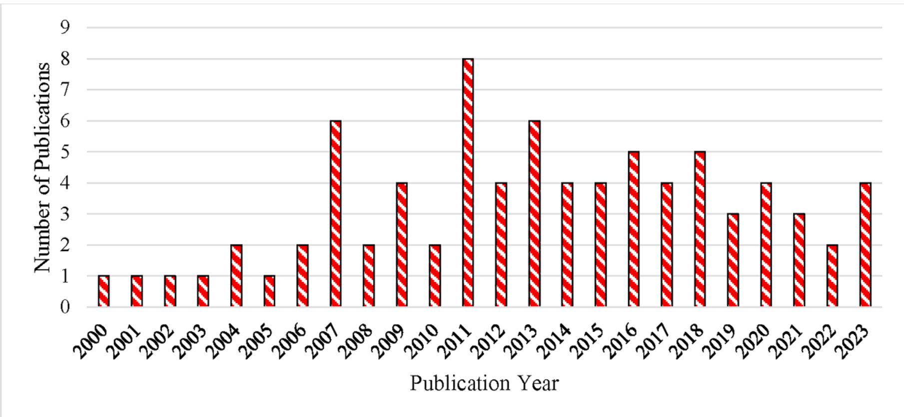

Fig. 5. Distribution of articles reviewed as per publication year.

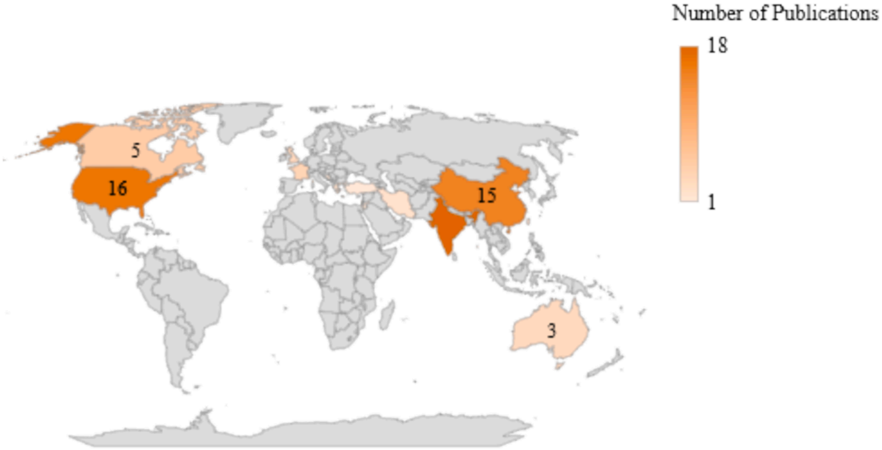

Fig. 6. Distribution of articles reviewed as per geographical area.

By addressing these limitations, the current study aims to highlight the challenges faced by reactive approaches, such as under-reporting, inconsistent data quality, and a lack of specialized accident investigation units in many developing regions. These issues underscore the need for proactive approaches, which are less reliant on historical crash data and instead focus on identifying and mitigating potential risks through behavioural observations and surrogate safety measures. The following sub-section reviews pedestrian level and site-specific factors in assessing safety in a proactive approach.

3.1. Pedestrian attributes

Pedestrian behaviour while crossing the road is widely considered an influential parameter for the assessing safety in a proactive approach. Pedestrian behaviour and attitude are dynamic in nature and often change according to their surroundings. As compared to drivers, pedestrians are not much constrained by regulations, are unpredictable and have more freedom to move and decide their path (Ren et al., 2011). This often results in frequent violations by pedestrians, especially at signalized intersections. Many of these pedestrian violations can be observed while crossing the road, such as crossing on red and not crossing along the crosswalk. Pedestrian's illegal crossing or violation behaviour could be considered as a substitute approach for assessment of the detrimental performance of urban signalized intersections (Koh & Wong, 2014).Several studies have tried to understand the factors associated with pedestrian's violation behaviour and risk-taking tendency. Pedestrian personal attributes include demography, socio-economy, crossing behaviours, convenience and safety perceptions. The succeeding sub-sections explores the various pedestrian attributes that are affecting safety and road crossing behaviours.

3.1.1. Pedestrian demographics

Pedestrian demography such as gender and age has been considered in several studies. Male pedestrians have a higher violation and risk-taking tendencies than females. Males also have higher signal violation frequency. Contrary to this, female pedestrians are more sensitive to risk perception and prefer to cross the road during the vehicle red phase and at designated crosswalks (Aghabayk et al., 2021; Brosseau et al., 2013; Díaz, 2002; Guo et al., 2011, 2012; Hamed, 2001; Haque & Kidwai, 2022; Marisamynathan & Perumal, 2014; Mukherjee & Mitra, 2020a; Rosenbloom, 2009; Tiwari et al., 2007; Tom & Granié, 2011; Yagil, 2000). Contrary to the above, a study from China found that male pedestrians are more likely to comply with traffic rules when crossing streets. The reason may be that some traffic and built environment characteristics might be dissimilar from previous studies (Ren et al., 2011). In several studies it has been reported that gender does not have a significant effect on pedestrian's compliance behaviour (Demiroz et al., 2015; Dommes et al., 2015; Yang et al., 2015; Zhang et al., 2016).

Pedestrian age had been reported as a significant predictor of crossing behaviour in several studies. Ren et al. (2011) collected observational based data from three cities in China and reported that pedestrians aged 60 and above were found to be most compliant with crosswalks and signals. Brosseau et al. (2013) reported that young adults in the age group of 18–35 have a higher violation tendency in Canada. Mukherjee and Mitra, (2017) performed an observational study in Kolkata, India. The authors observed that signal violation frequency is higher for the age group 16–49 and less for the age group above 50. Aghabayk et al. (2021) reported that older pedestrians of age over 61 years have lower odds of crossing during don't walk sign as compared to young pedestrians in Tehran, Iran. Several other studies also reported age as a significant factor in road-crossing decisions (Avineri et al., 2012; Demiroz et al., 2015; Guo et al., 2011; Oxley et al., 2005; Zhuang & Wu, 2011). The above findings conclude that old age pedestrians are associated with an increased level of compliance and law abidance. Older pedestrians suffer from reduced mobility, sensory and cognitive skills and thus prefer to wait longer and cross slowly only when safer crossing opportunities are available.

However, several studies have reported that does not have any significant impact on pedestrian violation behaviour (Dommes et al., 2015; Marisamynathan & Perumal, 2014; Mukherjee & Mitra, 2020c; Russo et al., 2018; Yang et al., 2015). Dommes et al. (2015)

Table 1
Summary of review – Proactive Approach.

S. No.	Authors	Study Area	Data	Dependent Variable	Independent Variables	Analytical Method
1	Yagil (2000)	Israel	Respondents: 203	Behaviour at signalized intersections	Gender, traffic volume, weather conditions	Descriptive, MLR
2	Hamed (2001)	Amman, Jordan	10 mid-block, Respondents: 350, Observed: 400	Waiting time, Crossing behaviour	Gender, Trip purpose, Traffic volume, vehicle type and speed	Maximum likelihood estimate, Risk function, Poisson Regression
3	Díaz (2002)	Santiago, Chile	Respondents: 146	Violation behaviour	Gender, Age	TPB, Descriptive, SEM
4	Oxley et al. (2005)	Australia	Respondents: 54	Gap selection, safety margin	Age, vehicle speed, time gap	Virtual simulation, hierarchical logistic regression,
5	Yang et al. (2006)	China	Respondents: 500, Video observations	Violation behaviour	Traffic volume, waiting time, weather	Micro-simulation, maximum likelihood estimate
6	Lobjois and Cavallo (2007)	Arcueil, France	Respondents: 78	Gap selection, safety margin	Age, vehicle speed	Virtual simulation
7	Arhin and Noel (2007)	Washington, DC, USA	Respondents: 102 and Video observations, 14 signalized intersections	Crossing and violation behaviour	Pedestrian countdown signal (PCS)	Descriptive, hypothesis testing
8	Hatfield and Murphy (2007)	Sydney, Australia	Manual observations: 546, 3 signalized and 3 un-signalized crossings	Crossing behaviour-mobile	Gender, crossing speed, waiting time	BLR, MLR
9	Van Houten et al. (2007)	Florida, USA	Manual observations: 2160, two signalized mid-block	Violation behaviour	Waiting time (vehicle green time)	Descriptive
10	D. Akin and Sisiopiku (2007)	Michigan, USA	Video observations,1-km section-5 signalized crossings,	Crosswalk and signal compliance	Pedestrian and vehicle volume, pedestrian signal, waiting time	MLR
11	Tiwari et al. (2007)	New Delhi, India	Video observations, 7 signalized intersections	Crossing behaviour	waiting time, Gender, pedestrian delay	Survival analysis, Kaplan-Meier survival curves
12	Nasar et al. (2008)	USA	Participants: 60 and Manual observations: 127, 3 intersections	Crossing behaviour-mobile	Situational awareness	Descriptive, hypothesis testing
13	Cambon de Lavalette et al. (2009)	Montreal, Canada	Manual observations: 4000, 12 signalized intersections	Violation behaviour	Pedestrian signals, number of lanes, traffic flow direction, refuge island	Galois Lattice, descriptive
14	Zhou et al. (2009)	Beijing, China	Respondents: 426	Road crossing decisions	Age, conformity tendency, self-identity, attitude	TPB, K-means cluster analysis, hierarchical MLR
15	Rosenbloom (2009)	Tel Aviv, Israel	Manual observations: 1392, 1 signalized intersection	Violation behaviour	Gender, group size, other people's behaviour,	Logistic regression
16	Faria et al. (2010)	Leeds, UK	Video observations: 365, 1 road crossing	Crossing behaviour	Gender, group size, nearest neighbour behaviour, social information	Simulation, Monte-Carlo analysis
17	Li and Fernie (2010)	Toronto, Canada	Video observations: 618, 1 signalized intersection	Violation behaviour	Refuge island, two stage crossing, walking speed, weather conditions	Descriptive
18	Cinnamon et al. (2011)	Vancouver, Canada	Manual observations: 9808, 7 signalized intersection	Violation behaviour	Built environment, land use, transit station, intersection geometry	Descriptive, GIS
19	Ren et al.(2011)	3 Cities, China	Respondents: 598 and Video observations: 6628 26 signalized intersections	Violation behaviour	Gender, age, group size, crosswalk length, signal timing, police presence	Odds Ratio, descriptive
20	Tom and Granié (2011)	Paris and Rouen, France	Manual observations: 400, 2 signalized and 2 un-signalized crossings	Crossing behaviour	Gender, on street parking, gaze pattern	Contingency table, descriptive
21	Wang et al. (2011)	Beijing, China	Respondents: 475 and Video observations: 1391 5 signalized intersections	Crossing behaviour	Waiting time, gender, age	Reliability and hazard analysis, exponential, log– logistic and Weibull distributions
22	Stavrinos et al. (2011)	USA	Participants: 108 (Exp. 1) and 59 (Exp. 2)	Crossing behaviour	Mobile use, gender	Virtual simulated environment, Bivariate correlations, simple contrast analysis
23	Guo et al. (2012)	Beijing, China	Video observations and questionnaire:1391, 5 signalized crosswalks	Crossing behaviour	Gender, age, waiting time, group size, traffic volume, trip purpose	Reliability analysis, log–logistic, non-parametric

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Table 1 (*continued*)

S. No.	Authors	Study Area	Data	Dependent Variable	Independent Variables	Analytical Method
						and extreme value distributions
24	Avineri et al. (2012)	Tel-Aviv, Israel	Video and questionnaire observations: 203, 1 signalized and 1 un-signalized crossing	Crossing behaviour	Age, gender, crossing speed, fear of falling	MLR
25	Cooper et al. (2012)	San Francisco, USA	Manual observations: 1144, 12 signalized intersections	Violation behaviour	Mobile use, gender, age, group size	Exploratory analysis
26	Schwebel et al. (2012)	USA	Participants: 138	Crossing behaviour	Mobile use	Virtual street environment, MLR, BLR
27	Thompson et al. (2013)	Seattle, USA	Manual observations: 1102, 20 signalized intersections	Crossing behaviour	Gender, mobile use, crossing speed, group size	Linear and logistic regressions
28	Supernak et al. (2013)	San Diego, USA	Video observations:5000, 1 signalized intersection	Violation behaviour	Crossing length, traffic volume, PCS, crossing speed, gender, age	Descriptive, hypothesis testing
29	Yang and Sun (2013)	Shanghai, China	Integrated: 106,Video: 1818 questionnaire observations: 356, 6 signalized intersections	Violation behaviour	Waiting time, signal timings, vehicle speed, crossing distance, time gap,	BLR
30	Lipovac et al. (2013)	Doboj, Bosnia and Herzegovina	Video observations: 21310, 2 signalized intersections	Crossing behaviour	PCS, gender, traffic volume, age	Exploratory analysis
31	Brosseau et al. (2013)	Montreal, Canada	Manual observations: 1102, 13 signalized intersections	Crossing behaviour	PCS, gender, age group size, pedestrian volume, waiting time	BLR, video based trajectory validation
32	Liu and Tung (2014)	Yunlin, Taiwan	Participants: 32	Crossing behaviour	Age, time gap, vehicle speed, time of day	Virtual simulation, BLR
33	Zaki and Sayed (2014)	Vancouver, Canada	Automated video observations: 376, 1 signalized intersection	Violation behaviour	Automated detection is significantly comparable to manual	Automated detection of violation behaviour
34	Marisamynathan and Perumal (2014)	Mumbai, India	Video observations: 775, 3 signalized intersections	Crossing behaviour	Crossing speed, gender, group size	BLR, Descriptive
35	Demiroz et al. (2015)	Izmir, Turkey	Respondents: 231 and Video observations: 993, 4 overpass locations	Crossing behaviour	Age, vehicle and pedestrian speed, distance gap	Exploratory analysis
36	Dommes et al. (2015)	Lille, France	Video and questionnaire observations: 422, 15 signalized crosswalks	Crossing behaviour	Age, parked vehicles, group size, traffic density, running, crossing path	BLR, step wise regression
37	Yang et al. (2015)	Beijing, China	Video observations: 1181, 5 signalized intersections	Crossing behaviour	Waiting time, left-turn phase upon arrival, traffic volume, group size	BLR, exponential, Weibull, lognormal and log-logistic
38	Basch et al. (2015)	Manhattan, NYC, USA	Manual observations: 21760, 5 signalized intersections	Crossing behaviour	Mobile use	Exploratory analysis
39	Asaithambi et al. (2016)	Mangalore, India	Video observations before and after signal installation, 1 intersection	Crossing behaviour	Crossing speed, waiting time, gender, age, crossing path, vehicle type and speed, traffic volume, time gap	MLR, descriptive analysis
40	Zhang et al. (2016)	Guangdong, China	Government crash records at intersections Pedestrians: 4817	Violation behaviour	Age, job, light conditions	BLR
41	Gillette et al. (2016)	Texas, USA	Manual observations: 760, 3 signalized intersections	Crossing behaviour	Mobile use, waiting time, age, gender	MLR,BLR
42	Muley et al. (2017)	Doha, Qatar	Video observations: 235, 1 signalized intersection	Crossing behaviour	Gender, waiting time, crossing speed, gaps, group size, distraction	Descriptive analysis, hypothesis testing
43	Lin and Huang (2017)	Tainan City, Taiwan	Participants: 24	Crossing behaviour	Mobile, situational awareness	Semi virtual environment, Descriptive analysis, hypothesis testing
44	Russo et al. (2018)	New York and Arizona, USA	Video observations: 3038, 4 signalized intersection	Crossing behaviour	Mobile use, crossing speed, group size	BLR
45	Jiang et al. (2018)	China	Participants- Video and eye tracker observations: 28, 1 signalized intersection	Crossing behaviour	Mobile use	Descriptive analysis, hypothesis testing
46	Mukherjee and Mitra (2020a)	Kolkata, India	Respondents: 3250, Video observations: 65500, 55 signalized intersections	Violation behaviour	Marked crosswalk, street parking, cycle length, land use, gender, carrying	Beta regression

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Table 1 (*continued*)

S. No.	Authors	Study Area	Data	Dependent Variable	Independent Variables	Analytical Method
					overhead load, student, perceived satisfaction	
47	Zhang et al. (2020)	Florida, USA	Video observations: 122, 1 signalized crosswalk	Violation behaviour	Gender, crossing direction, group size, location	Automated detection of violation behaviour, ANN-LSTM
48	Aghabayk et al. (2021)	Tehran, Iran	Manual observations: 552, 2 signalized and 2 un-signalized intersections	Crossing behaviour	Gender, age, mobile use, crossing speed, group size	MLR,BLR
49	Zhang et al. (2021)	Florida, USA	Video observations: 182, 3 signalized intersections	Violation behaviour	Pose estimate, walking speed, waiting time, green time	Automated detection of violation behaviour using pose estimates, ANN- SVM, RF, GBM, XGBoost
50	Tian et al. (2022)	Leeds, UK	Simulation, Participants: 60	Crossing behaviour	Gap acceptance, safety margin, vehicle speed	Simulation, Linear and logistic regression
51	Sheykhfard et al. (2023)	Babol City, Iran	Video observations: 629, 2 signalized intersections	Violation behaviour	Pedestrian, road, vehicle and environmental factors	Structural equation modelling, K-mean clustering

MLR-Multinomial Logistic Regression; TBB-Theory of Planned Behaviour; SEM-Structural Equation Modelling; BLR-Binomial Logistic Regression; GIS-Geographical Information System; ANN-Artificial Neural Network; LSTM-Long Short Term Memory; SVM-Support Vector Machine; RF-Random Forest; GBM-Gradient Boosting; XGBoost-Extreme Gradient Boosting.

reported that age is not associated with signal violation but older pedestrians were observed to exert more cautious behaviour while crossing the road. Yang et al. (2015) concluded that age is an insignificant predictor but observed a slightly higher compliance in younger and slightly lower in older pedestrians. The behaviour of young pedestrians was attributed to the presence of high schools and campuses in the survey area whereas older pedestrians' behaviour to past crossing habits. So, age alone cannot explain why pedestrians violate or do not violate the signal, but it manipulates how pedestrians behave. Further comfort and convenience play a vital role in road-crossing decisions (Haque & Kidwai, 2023a).

3.1.2. Group size

Situational variables, such as group size have a significant impact on pedestrian compliance. Several studies have shown that the level of pedestrian density (group size) crossing or waiting together significantly affects signal infringement behaviour. Studies indicate a clear trend that as the group size increases the signal violation behaviour decreases (Brosseau et al., 2013; Dommes et al., 2015; Guo et al., 2011; Mukherjee & Mitra, 2017; Rosenbloom, 2009; Sheykhfard et al., 2023; Yang et al., 2015; Zhuang & Wu, 2011). Single pedestrian are more non-compliant than larger groups, due to freedom from platoon and directional effects (Marisamynathan & Perumal, 2014).As pedestrians wait longer at an intersection, more people join them, increasing the group size. A larger group size makes violation behaviour less likely. Such behaviour can be explained on the basis of conformity psychology. Similar conformity tendency behaviour has been reported in many studies (Faria et al., 2010; Yagil, 2000; Zhou et al., 2009). Few studies have reported contradictory results that signal compliance rate, for pedestrians walking in groups are low as compared to crossing alone (Haque & Kidwai, 2021; Ren et al., 2011). Further pedestrians in groups of 5 or more are more likely to commit signal violations (Russo et al., 2018). The authors explain that pedestrians in a group are more irresponsible regarding the potential risks.

3.1.3. Other pedestrian behaviour

Yagil (2000) conducted a self-reported questionnaire survey and found that the presence of other pedestrians affects pedestrian road crossing behaviour. The presence of children and other pedestrians waiting for a walk signal increases the likelihood of others waiting for the pedestrian green signal. Furthermore, women, as compared to men are more affected by the presence and behaviour of others. Contrary to above, it was found that males tended to follow others more than females (Yagil, 2000).

Yang et al. (2006) found that the majority of pedestrians have medium to high tendency of signal violation if they see others doing the same. Zhou et al. (2009) used the theory of planned behaviour (TPB) and found that pedestrians were more likely to cross the road against red light when other pedestrians crossed as compared to the situation when others waited. Faria et al. (2010) used an observation based study to analyse the impact of social information on road crossing behaviour. The result shows that pedestrians were 1.5–2.5 times more likely to cross if one of their neighbours had already begun to cross irrespective of the state of signal. The result is contrary to previous result which reported that pedestrians are more influenced by other's behaviours during don't walk signal phase (Rosenbloom, 2009). Furthermore, it was found that males tended to follow others more than females which is in contrast with previous result (Yagil, 2000).

The psychology and beliefs behind following or not following are well explained by (Yagil, 2000). Pedestrians who do not cross during red phase are found to be influential and apparently affect road crossing tendency by stimulating conformity. Crossing the road while others wait for green phase is likely to be professed as violation by fellow pedestrians and might portray a negative image. While abstaining from crossing during red phase when other are crossing, might be interpreted as pedestrian's personal principles and moral behaviour. The other explanation might be that crossing while other pedestrians wait implies being watched from the back and might generate uncomfortable feelings, on the other hand watching others is more likely a psychologically unbiased feeling.

3.1.4. Technological distraction – Mobile phone usage

In recent years, there has been an exponential increase in mobile phone users, particularly smartphone users, worldwide. Although mobile phones are a convenient choice for communication, they result in inadvertent traffic safety issues due to the distractions they cause. Technological distraction includes behaviours such as talking, texting on a mobile phone, using headphones, reading and browsing or simply looking at a mobile phone. The safety consequences of distracted driving are well recognized but the potential impacts of distracted walking require more focus and attention. Nevertheless, several researchers have tried to study the behaviour of distracted pedestrians.

Distracted pedestrians are less likely to show safe crossing behaviour (Haque et al., 2024). Hatfield & Murphy (2007) reported that the use of mobile phone is highly related to gender-specific unsafe crossing behaviour. The authors concluded that females using mobile phones are less likely to look at traffic before and during crossing and also less likely to wait for traffic to stop. Russo et al. (2018) conducted a very comprehensive study to correlate pedestrian demographic factors with violation and distracted walking behaviours. The authors reported that females, crossing alone and those who commit crosswalk violations are more likely to talk on phone while walking. Cooper et al. (2012) reported that mobile usage among pedestrians while road crossing varied from 3 % to 18 % at specific study sites. Females, young adults (18–34 yrs.) and pedestrians crossing individually are more likely to use mobile phones while crossing the road. Another observational study conducted in Seattle, USA, reported that nearly 30 % of pedestrians were distracted while crossing. Further, it affects other types of unsafe behaviours such as not looking left and right before crossing.

Among distracted behaviours, texting has the highest odds of showing unsafe behaviour (Thompson et al., 2013). Basch et al. (2015) performed one of the largest studies to understand technologically distracted walking behaviour and found that, 42 % of the total signal violating pedestrian were distracted. Gillette et al. (2016) performed a similar observational and reported that, 28 % of the pedestrians were technologically distracted with a large share of them texting. Jiang et al. (2018) performed a study to understand the effect of mobile phone usage on crossing behaviour at signalized intersections and found that text distraction has highest levels whereas music distraction has lowest level of impairment on crossing performance of pedestrians. In another study conducted in Tehran, Iran, the result show that 44.6 % of pedestrians were technologically distracted. Further, pedestrians, texting are less likely to pay regular attention to the traffic while crossing the road at signalized intersections (Aghabayk et al., 2021).

In a simulation-based study it was found that pedestrian texting or listening to music were more likely to be hit by moving cars while crossing as compared to undistracted pedestrians (Schwebel et al., 2012). Similar to the above study, another study conducted in semi-virtual environment concluded that distracted pedestrians have delayed responses to detect and react to roadside events while reading and texting on smart phones (Lin & Huang, 2017). Mobile phone use while crossing is associated with reduced situational awareness and increased unsafe crossings and might increase the risk of accidents (Nasar et al., 2008). This is further supported by another simulation based study that reported, cell phone usage while crossing the road might enhance unsafe pedestrian behaviour among college students to a level of compromised safety (Stavrinos et al., 2011).

In light of the above results several important conclusions can be made in relation to technologically distracted walking behaviour. Use of electronic devices while walking causes certain cognitive and sensory impairments resulting in unsafe crossing behaviours. Distracted persons have less situational awareness and are more likely to compromise their safety in traffic. Females, young adults and crossing alone have higher odds of using electronic devices and exhibiting unsafe crossing behaviours. Among the various types of technological distractions texting is associated with the most while listening to music via headphones has the least impairment and unsafe crossing behaviours. Finally distracted walking significantly reduces the safety and efficiency of transportation systems.

3.1.5. Crossing speed

Pedestrian crossing speed is a crucial parameter for designing of pedestrian facilities. It determines the time required by pedestrians to safely cross the road. It is found to vary largely between 1.2 and 1.4 m/s. The existing manual of the Indian Road Congress (IRC) considers walking speed at zebra crossing as 1.2 m/s (IRC 93, 1985). The Highway Capacity Manual recommends using a walking speed of 1.2 m/s if elderly pedestrians comprise less than 20 % of the population, and 1.0 m/s if they comprise more than 20 % (HCM, 2000). This steady value is not relevant for dynamic traffic environments existing at signalized intersections. Pedestrian crossing speed is highly subjective to the pedestrian's demographic and crossing behaviour. Zaki and Sayed (2014) performed an automated analysis of pedestrians' violation behaviour at urban signalized intersections and found that a significant difference exists in average crossing speed between violating and non-violating pedestrians.

Gender, age group and group crossings have a significant effect on crossing speed at both signalized and un-signalized locations. Marisamynathan and Perumal (2014) reported that the mean crossing speed of pedestrians was 1.34 m/s and it varies across gender, age group and group size. It was found that males and young pedestrians have significantly higher crossing speed as compared to their counterparts (Aghabayk et al., 2021; Asaithambi et al., 2016; Marisamynathan & Perumal, 2014; Muley et al., 2017).Contrary to above results, a simulation based study found that elderly people tend to walk quickly or frequently chose to run across the road while young people were more likely to cross the road at their normal speed (Liu & Tung, 2014). Crossing speed tends to decrease with an increase in group size due to platoon and directional effects (Marisamynathan & Perumal, 2014; Muley et al., 2017). Similar results were obtained by another study which concluded that at un-signalized locations group crossing is slower whereas at signalized locations it is faster (Aghabayk et al., 2021). Pedestrians crossing perpendicularly have higher crossing speeds than their counterparts (Asaithambi et al., 2016).

Crossing speed is also significantly related to pedestrian violation behaviours. Crossing speed during the red phase (violating pedestrians) is higher than during green phase. Pedestrians showing spatial or temporal violations are much faster as compared to others (Aghabayk et al., 2021; Haque & Ahmad Kidwai, 2023). The results are similar to prior research in this field (Russo et al., 2018). It is due to the fact that during the violation phase, pedestrians cross the road abruptly at higher speed in order to avoid conflicts with vehicles.

Pedestrian crossing speed is also influenced by the pedestrian's state during crossing, such as physical and technological distraction. Non-distracted pedestrians have higher crossing speed than distracted pedestrians (Muley et al., 2017). Russo et al. (2018) reported that talking or texting on mobile is negatively but trivially related to walking speed. Pedestrians using headphones exhibit faster whereas those in other distraction groups have slower walking speeds than undistracted pedestrians. Various other studies have also established the fact that talking or texting while crossing is associated with slower crossing speed (Hatfield & Murphy, 2007; Thompson et al., 2013).

3.1.6. *Waiting time*

Pedestrian waiting time is an influential parameter in road crossing behaviour. Once pedestrians reach their threshold value of waiting time, they often become impatient and exhibit dangerous behaviour by crossing against the red light. Highway Capacity Manual (2000) concludes that a waiting time of more than 30 s makes pedestrians impatient and intolerant, compelling them to be involved in illegal crossing. Indo-HCM (2017) recommends the average waiting time for males from 5.27 to 8.38 s whereas for females it is from 5.65 to 10.54 s. Wang et al. (2011) conducted a study in Beijing and found that, the longer pedestrians wait for crossing, the greater their probability to cross illegally. Similar results have been obtained in studies conducted in other countries (Brosseau et al., 2013; Guo et al., 2012; Tiwari et al., 2007; Yang et al., 2015). A 10 % increase in mean waiting time will enhance the probability of violation by 7.9 % (Brosseau et al., 2013). Further, majority of pedestrians cannot wait for more than 40 s. The waiting time for risky pedestrian is found to be 1.2–3 s and for those pedestrians who are willing to wait for safer crossing opportunities is 50 s (Guo et al., 2012; Haque & Kidwai, 2023b). Another research also suggested limiting the minimum green time of vehicles to 30 or 60 s, so that the pedestrians do not face higher delays and become impatient (Van Houten et al., 2007).

Pedestrian personal traits such as age and gender were also found to have a significant impact on waiting duration. Waiting time for females is found to be 27 % more than males. Haq et al. (2024) concluded in his study that males perform riskier crossing than females at unsignalized midblock crosswalks. Furthermore, about 84 % of pedestrians were found to have waiting time of less than 10 s (Tiwari et al., 2007). Similar results were obtained at mid-block locations as well (Van Houten et al., 2007). Liu & Tung (2014) performed a simulation based study to assess pedestrians' road crossing decisions and found that pedestrians have higher tendency to wait for signals in darkness. Yang et al. (2015) performed an observational study to investigate pedestrian crossing behaviour and waiting time in Beijing, China. Two types of pedestrian behaviour are identified according to waiting time- immediate crossing behaviour and waiting behaviour. About 14.3 % of pedestrians showed immediate crossing behaviour or violation behaviour. Further left-turn phase upon arrival, and traffic volume had considerable effects on the odds of pedestrians' immediate crossing behaviour. In a recent study Mukherjee & Mitra (2020c) made a similar conclusion that an increase in signal cycle length urges pedestrians to perform unsafe crossing.

Based on the above discussions, it can be concluded that the probability of pedestrians making unsafe crossings varies with waiting time. With an increase in the waiting time, pedestrians get impatient and violate the traffic signal. Further waiting time should be designed to accommodate the needs and comfort of pedestrians. Wherever possible, the waiting time should not exceed 60 s. The provision for all red time for vehicles can also be an effective solution to counter pedestrian violation behaviour due to increased waiting time.

3.1.7. *Trip purpose*

Pedestrian trip purpose is also associated with compliance behaviour. Several studies have found that trip purposes related to work or school are significantly linked to higher violation and reduced waiting tendency (Guo et al., 2011, 2012; Hamed, 2001; Marisamynathan & Vedagiri, 2018; Mukherjee & Mitra, 2020a; Wang et al., 2011). This is because these pedestrians do not want to be late and, therefore, try to minimize their travel time to their destinations.

3.2. *Site specific factors*

Site-specific factors significantly influence pedestrians' safe crossing behaviours and decision-making on road crossings. Numerous studies have examined how the characteristics of existing site conditions shape pedestrian behaviour, highlighting the intricate relationship between the environment and human actions. These factors encompass a broad range of elements, including the built environment, infrastructure quality, roadway geometry, traffic characteristics, and the operational and control systems in place. This section delves into the impact of various site-specific factors on pedestrian road-crossing decisions.

3.2.1. *Built environment – Land use*

Built environment is one the most crucial factor affecting pedestrian crossing behaviour at signalized intersections. It is found that the inconsistencies in pedestrian violations can be attributed to the characteristics of locations. Certain types of land use, such as commercial, industrial and alcohol outlets might influence pedestrian to cross against the signal (Cinnamon et al., 2011). The presence of a public transit hub increases violations as pedestrians try to avoid missing a transit connection. Interestingly, the presence of slum population in the vicinity of intersections also tends to increase the violation behaviour (Mukherjee & Mitra, 2020a). Contrary to the above findings presence of offices and educational institutes reduces violations at reasonable rates (Guo et al., 2011). Muley et al. (2017) concluded that unsafe pedestrian crossings are highly influenced by built environment features at the site.

3.2.2. Intersection characteristics

The effects of presence of median, refuge island, type and width of refuge on pedestrian unsafe behaviour have been considered in a few studies. Ni et al. (2017) and Faizan ul Haq et al. (2024) reported that the presence of refuge island significantly reduces the violation tendencies and increases the safety perception. A considerable number of studies tried to study the influence of crossing length or crosswalk width on pedestrian crossing decisions. Crossing length and interfering traffic volume were found to be the most important predictors of signal violation. It was found that the proportions of violations are substantially less on longer crossings as compared to shorter crossings (Supernak et al., 2013). Yang and Sun (2013) concluded that crossing distance is negatively associated with violation behaviour suggesting that a short crossing distance will increase the chance of pedestrian red-time crossing. Ren et al. (2011) also found that compliance is low at shorter (8–11 m) and longer (18–23 m) crossing distances. The study explained that at shorter crossing distances, pedestrians become ignorant of risk and confident in finding suitable gaps to cross the road safely. At longer crossing distances they might end up showing violations. Contrary to the above results, it was found that violation increases proportionally with increase in number of lanes from one to three (Cambon de Lavalette et al., 2009).

Another important location specific factor that notably affects the safe crossing behaviour of pedestrians is presence of properly marked crosswalks at signalized intersections. In an early study, Ekman and Hyden (1999) reported that, in Scandinavian countries, the absence of crosswalks significantly increases pedestrian dangerous behaviours. The safety potential is great at properly marked zebra crossings as majority of crossings occur at these locations. In another, it was found that more than 80 % of pedestrians were crossing the road along the crosswalk whereas only about 50 % obeyed signals. The study concluded that signalized intersections with properly marked crosswalks are very attractive and well-identified site by pedestrians to cross the road (Akin & Sisiopiku, 2007). Rankavat and Tiwari (2016) found that crosswalk is the most preferred choice for crossing the road because of its perceived convenience. In a recent study, it was found that the presence of crosswalks reduced the proportion of signal violations by 69 % (Mukherjee & Mitra, 2020c). Sisiopiku and Akin (2003) concluded that signalized intersections with zebra crossings help to channelize the flow of pedestrian crossings and its absence would result in pedestrian crossings spread out all over the intersection shared space; however, the presence of crosswalks was unable to convince pedestrians to comply with the signal particularly under low traffic. Properly designed and placed pedestrian facilities encourage users to cross at specific locations.

3.2.3. Pedestrian countdown signal (PCS)

The effect of pedestrian countdown signal (PCS) on crossing violations has also been considered in a few studies. Studies have found that the presence of PCS reduces the pedestrian violation tendency. Lipovac et al. (2013) concluded that a statistically significant signal infringement occurred at pedestrian crossings without PCS rather than at crossings with countdown displays. The presence of PCS effectively reduces female violations in particular. The presence of countdown display is negatively linked with the probability of violations. At intersection crosswalks with PCS, the inclination of infringement or dangerous crossing is about 15 % lesser than at locations without a countdown display (Brosseau et al. 2013). Arhin & Noel (2007) found that pedestrians are collectively convinced about their increased perception of safety due to the presence of the PCS. In another study it was found that PCS is effective in reducing violations at long crossings and is rather ineffective at short crossings (Supernak et al., 2013).

3.2.4. On street parking

Studies have also focused on the connection between the presence of on-street parking near the intersection and pedestrians' violation behaviour. Presence of on-street parking has been associated with a 30 % increase in violation behaviour (Mukherjee & Mitra, 2020a). A study conducted in France found that the presence of on-street parking was associated with an increase in signal violation probability. The authors suggested that the presence of parked vehicles block pedestrian visibility of ground, traffic, other pedestrians and signals, which may result in violations (Dommes et al., 2015). Another study presented a similar conclusion regarding the presence of on-street parking (Tom & Granié, 2011).

Yannis et al. (2013) had entirely conflicting conclusions related to the effect of illegal parking on pedestrians' crossing decisions in Athens, Greece. The study concluded that the presence of illegally parked vehicles in the vicinity of the crossing locations obstructs visibility and hence pedestrians became more cautious, leading to fewer violations. Dommes et al. (2015) attributed the contrast in results to crossing situations. In case of controlled situations, such as signalized intersections, the presence of parked vehicles might augment pedestrians' safety perception (by decreasing road width) and cause signal violation. While in uncontrolled situations at mid-blocks pedestrians would observe much more cautious behaviour.

3.2.5. Traffic

Traffic operational factors, such as vehicle volume, speed and available gaps are found to have pertinent effect on pedestrians' road crossing behaviour. Several studies have found that an increase in traffic volume is associated with less frequent violations. Yagil (2000) tried to understand the beliefs, motives and situational factors related to pedestrian crossing behaviour at signalized intersections. The author concluded that a high traffic volume is associated with an increase in waiting tendency whereas low traffic volume increases the signal violation tendency. Approaching traffic volume, vehicle speed and type (LMV/HMV) are influential in determining the pedestrian's waiting time (delay) and the number of crossing attempts. Increase in traffic volume and speed increases the waiting time due to the availability of fewer gaps. Further if the oncoming vehicle is HMV the pedestrian becomes cautious and aborts crossing attempts and waits additionally (Hamed, 2001). A micro-simulation based study indicates that low traffic volume and long duration of red signal phase contributed more to the odds of pedestrian signal violation (Yang et al., 2006). Guo et al. (2011) used hazard based duration approach and applied Relative Hazard Ratio (RHR) to assess and model pedestrian behaviour. It is found that the increase in traffic volume decrease the RHR hence decreasing the probability of violations. Similar results were obtained in other studies as well (Guo et al., 2012; Wang et al., 2011). Generally, under low traffic volume pedestrians have a higher possibility of violation behaviour than those under medium and high volume conditions. This is because under high traffic volume the available gaps between vehicles decrease and chances to cross the road are relatively unsafe (Yang et al., 2015).

When a vehicle approaches, the pedestrian's decision to cross a road should preferably include time gap, distance between the pedestrian and oncoming vehicle, vehicle speed and the personal mobility of the pedestrian (Liu & Tung, 2014). Several empirical studies have reported that pedestrian decision to cross mainly depend upon distance perception rather than speed perception (Oxley et al., 2006; Oxley et al., 2005). The reason is the inability of pedestrians to accurately judge the speed of vehicles whereas the distance gap can be judged with a fair degree of accuracy as it can be visualized with reference to some distant objects. For instance 63 % of pedestrians rely on the distance between the approaching vehicle and themselves and only 10 % on the speed of the approaching cars for making their crossing decisions (Connelly et al., 1998). Lobjois and Cavallo (2007) performed a simulation-based study to understand the effect of age on road crossing decisions considering the speed of approaching vehicles. It was found that elderly people are more likely to make their decisions based on distance gap rather than vehicle speed.

Studies have applied the concept of safety margin to judge pedestrians' road crossing decisions. The smaller the safety margin, the larger the risk. In general, the safety margin reduces with an increase in road crossing time, as pedestrian spends more time on the road. The safety margin for the elderly is lower than that for the young pedestrians, because of their reduced mobility. When the speed of an oncoming vehicle is high, elderly pedestrians are more likely to make unsafe decisions than when the oncoming vehicle is slow (Lobjois & Cavallo, 2007; Oxley et al., 2006; Oxley et al., 2005; Tian et al., 2022). Speed and distance of conflicting vehicles also have a considerable effect on pedestrians' road crossing decisions (Haq et al., 2024). Yang and Sun (2013) obtained a contradictory result and reported that signal violation increases with an increase in approaching vehicle speed and available time gap. This indicates that pedestrians may fail to sense the speed of passing vehicles and the result may be severe personal injury. This is because, although pedestrians are aware that sufficient contemplation should be given to vehicle speed while crossing the road, but they are neither sensitive nor perceptive of the changes in vehicle speed. The effect is more prominent in elderly pedestrians, due to lack of agility and mobility (Liu & Tung, 2014).

3.2.6. Temporal factors

Temporal factors, such as time of day and weather also have been found to affect pedestrian behaviour. Pedestrians are found to wait longer for traffic signal in darkness (Liu & Tung, 2014; Yagil, 2000). Zhang et al. (2016) found that the risk of pedestrian signal violations is higher under relatively good lighting conditions, during bad weather and on weekdays. Similar results were reported in other studies, that bad weather such as snowfall or rain, decreases the tendency to wait for walk signal phase (Li & Fernie, 2010; Yagil, 2000). Contrary to the above results, the probability of pedestrian signal non-compliance is found to be low to very low during rainy or snowy conditions (Yang et al., 2006).

4. Conclusions

Globally, road accidents involving pedestrians are raising concerns. A significant number of research studies have been conducted to address this issue and improve pedestrian safety. The available state of the art approaches to studying pedestrian safety issues can be broadly categorized into reactive, proactive and pedestrian perception-based approaches. A reactive approach is primarily suited for developed countries where crash records are consistently available with a high degree of accuracy and reliability. Reactive approach cannot be effectively used in developing economies as it suffers from several shortcomings related to data, such as underreporting, missing details, accuracy and reliability. On the contrary, proactive approach has been suggested by researchers for developing countries because of its cost-effectiveness (Kadali & Vedagiri, 2016). Therefore, the current paper performed a systematic and comprehensive literature review, focussed on proactive approach, to investigate the developments in the field of pedestrian safety, particularly at urban intersections. The purposes of the review are to analyse the existing state of the art, identify existing problems and understand the factors affecting pedestrian behaviours and safety. The expositions, key findings, limitation and future scope are discussed below.

Based on the findings of this review, several strategies and interventions are proposed to improve pedestrian safety at intersections, focusing on both behavioural, infrastructure and operational factors. Pedestrian behaviour and demographic differences require targeted interventions to enhance safety. Educational campaigns should focus on high-risk groups such as young males, who exhibit higher tendencies for unsafe behaviours like signal violations. Infrastructure designs, such as extended crossing times and auditory signals, can benefit elderly pedestrians with reduced mobility. Group dynamics also play a significant role in crossing behaviour; the use of visual aids, such as pedestrian countdown signals, can improve compliance, especially in larger groups. Additionally, promoting staggered crossings in high-density areas can mitigate platoon effects and enhance visibility, thereby reducing unsafe crossing behaviours. Technological distractions, such as mobile phone use, pose significant risks. Creating "distraction-free zones" near pedestrian crossings, complemented by warning systems and public awareness campaigns, can help address this issue. These campaigns should particularly target younger pedestrians, who are more likely to engage in distracted walking. Installation of tactile surfaces near crossings would also help the distracted pedestrian to be cautious. Crossing speed and waiting time should also be optimized by employing dynamic signal systems that adjust based on real-time pedestrian density and traffic conditions.

The built environment significantly influences pedestrian behaviour. Infrastructure improvements, such as installing refuge islands and well-marked crosswalks, can guide pedestrian flow and reduce violations. Urban designs that prioritize shorter crossing distances and pedestrian-friendly layouts should be incorporated into planning. Smart technologies, such as adaptive signals and illuminated crosswalks, enhance pedestrian visibility and promote compliance, particularly at signalized intersections. Regular audits of pedestrian facilities are necessary to ensure their continued effectiveness. Traffic operational factors also warrant specific interventions. Speed calming measures near intersections can address the risks posed by high-speed traffic. Furthermore, the presence of law enforcement personnel or automated violation detection systems can encourage pedestrian compliance with signals and road rules. Pedestrian countdown signals (PCS) have shown significant promise in reducing violations and should be widely implemented, especially at intersections with longer crossing distances. Temporal factors, such as lighting and weather, must be considered in pedestrian safety strategies. Signal systems that are responsive to adverse weather conditions can improve comfort and safety for pedestrians. Enhanced lighting at intersections, particularly during night, ensures better visibility and reduces risks associated with low-light conditions.

These recommendations underscore the importance of a multi-faceted approach to pedestrian safety, combining behavioural interventions, infrastructure enhancements, and technological innovations to create safer intersections. By addressing the identified factors systematically, these strategies have the potential to significantly improve compliance and enhance pedestrian safety at urban intersections.

Several factors related to pedestrian attributes and site specific elements were found to have significant effect on pedestrian behaviour and compliance. However, contradictions have been observed in some of the factors assessed for this review. The differences in the findings can be attributed to several factors. Many studies relied on observational or self-reported data to analyse pedestrian behaviour, which are responsible for potential biases. Observational studies, while valuable for capturing real-world behaviours, can be limited by subjective interpretation and may miss subtle contextual factors, such as pedestrian distractions or environmental variables. In order to minimize these limitations, the use of automated data collection methods, such as video analysis and sensor-based tracking are recommended. Limited sampling sizes or short observation periods used by few studies may not capture the full range of complex pedestrian behaviours. Further, certain studies derive their findings from local socioeconomic and geographic contexts, which may not be applicable globally. This highlights the need for versatile studies to verify if observed behaviours and safety measures hold true across different economic and spatial contexts. Finally, the contradictions can be attributed to different statistical methods used to analyse pedestrian behaviour, suggesting the need of robust model validation techniques and models that balance complexity with interpretability.

The study provides a systematic analysis of proactive approaches to pedestrian safety at intersections, offering insights into pedestrian behaviour, site-specific risk factors, and effective intersection features. By focusing on proactive rather than reactive measures, the study contributes to a growing body of research advocating for early intervention strategies to reduce pedestrian risk. This is particularly valuable for developing countries where data limitations often restrict the use of reactive methods. The systematic review identifies the most influential factors—such as pedestrian demographics, behaviours, intersection elements and traffic characteristics—that can be used to predict and pre-emptively mitigate risks, enhancing safety even in data-scarce environments. Furthermore, the research also introduces a proactive framework that can be customized to different intersection and urban settings, providing insights for urban planners and policymakers.

Pedestrian behaviours are stochastic and have more freedom for their movement. Several factors related to pedestrian personal attributes and site specific are found to have significant and pertinent effect on their behaviour and safety. Therefore, region and area specific understanding is required, to suggest measures for improvement of pedestrian safety. Each intersection has unique characteristics and might require targeted engineering and behavioural interventions in order to reduce violations and accidents.

The current study primarily investigated the pedestrian behaviours and site specific factors, it does not fully account for the dynamic interactions between pedestrians and vehicles, which are crucial in intersection safety analysis. The study does not extensively address the role of newer technologies like automated pedestrian detection, adaptive signal controls, or AI-driven predictive models, which have shown potential in proactive safety measures. This study provides a critical review of current proactive approaches but lacks a longitudinal perspective to assess the long-term impact of proactive safety interventions. The reviewed literature includes studies from specific geographic regions, potentially limiting the applicability of findings to other locations with different infrastructure, cultural, or economic factors. Based on the above limitations, future studies could focus on research using simulation models or observational methods that include both pedestrian and vehicular behaviours. Additionally, the role of newer technologies with longitudinal research, can be explored. Finally, the study also suggests a need for comparative analyses across different economic and geographical contexts.

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Faizanul Haque: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Farhan Ahmad Kidwai:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Muhammad Faizan ul Haq:** Writing – review & editing, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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